

1. → Explain briefly with an example the necessary steps involved in a typical PR system. (10)

→ Assume 3 disjoint pattern class w_1, w_2 and w_3 in R^2 . Compute the decision boundaries and decision rules that are needed to successfully classify all the patterns from the said 3 classes for the following two cases —

(a) Each class is separable from all other class (6+6)

(b) Classes are pairwise separable (10)

2. → How binning and clustering can be used to handle noisy data. (5+5)

→ Explain briefly a typical Naive Bayes classifier. What are the advantages of such a classifier over KNN and Decision Tree? (6+4)

3. → What is partitional clustering? State and explain one such algorithm with example. Discuss the merits and demerits of the method. (2+12+6) → (based on the principle of minimization of intra class distances)

4. → Explain the following terms in connection to a SVM classifier (2+10)

(a) Maximum Margin Hyperplane

(b) Kernel Function

→ Suppose that two pattern classes are linearly separable. Describe a method using a suitable ANN that can successfully classify the said two patterns classes. (12)

5. Short Notes on — (10 each)

(a) Importance of Syntactic PR

(b) Evaluation of clustering results

(c) Syntactic Pattern Description.

(d) Bayesian Belief Network

(e) Template matching vs parsing in syntactic PR

(f) Importance of non-metric similarity measure in clustering.

(g) Hyperplane properties

(h) Kohonen's Self Organizing Network → (How such a network can be employed to solve some typical problems of PR)

(i) ISO DATA algorithm

What is syntactic PR? Why it's important? (2+2)

With necessary diagram explain briefly a typical Syntactic PR system. (8)

6x) Describe briefly how pattern classification can be implemented using decision function? (4)

→ Describe briefly the factors that affect the success of such a pattern classification scheme. (4)

7x) How are the features of a biological neurone imitated in an artificial neurone? (6)

→ Describe how can a single layer perceptron classify two linearly separable classes? Can such an ANN learn an XOR function? Justify (10+4)

8x) What are the expected features of a typical density-based clustering algorithm?

→ State and explain DBSCAN algorithm for clustering.

→ Discuss how a user can assign the values of the parameters used in DBSCAN algorithm.

→ What are the merits and demerits of the algorithm? (2+10+4+4)

9x) Suppose you need to classify four alphabets. Define necessary set of primitive strings and an appropriate grammar to solve the problem using syntactic pattern recognition method. (8)

10x) Design a classifier that can classify a set of given text documents into a no. of classes depending on their contents where the no. of such classes is known a priori. Comment on the merits and demerits of such a classifier. (12+2)

→ Explain briefly the necessary steps involved in a typical pattern classifier (6)

PR CT-2 [2023] + (8x6)=18

1. A node in a typical ANN does not usually have local memory. F.
 2. Syntactic PR attempts to classify patterns based on a set of extracted features and an underlying statistical model. T
 3. The single-layer perceptron cannot solve the X-OR problem. T
 4. A syntactic PR classifier infers a grammar from a set of training patterns. T
 5. Use of appropriate values of Tuning parameters is important for proper placement of support vectors in SVM.
 6. Use of a very large value of Gamma underfits a SVM model.
- 3.1. with necessary diagram explain briefly a typical syntactic PR system. (6)
2. How are the features of a biological neurone imitated in an artificial neurone. (6)
3. Importance of Kernel functions in SVM. (6)

TRUE/FALSE (2 Marks)

- 1) A naive Bayes classifier can only work on continuous valued data. (F) *categorical data*
- 2) The success of a pattern classification scheme using decision function depends only on finding the appropriate form of $d(x)$. (F) R3611
- 3) In a typical PR problem, the dimensionality of measurement space is always less than that of feature space. (F)
- 4) Missing data in a database may be due to the error of the equipment in recording the data. (T)
- 5) Data preprocessing is required to remove the noise if any from the data. (T)
- 6) In Bayesian network, the nodes represent events and the edges represent conditional dependencies. (T)
- 7) For a supervised pattern classification problem having M classes, where the patterns are pair-wise separable, the classifier needs to compute $M/2$ no. of decision surfaces to perform classification. (F)
- 8) Data transformation methods provide a subset of the original data. (T)
- 9) The merit of a NB classifier is that, the assumption of class conditional independence is always valid for real life problems. (F)
- 10) Binning is a method of data preprocessing to detect and remove outliers.
- 11) ~~Spatial~~ Each node in a typical ANN can have any no. of outgoing connections, where the signals in all of these may not be the same. (F)
- 12) The Tamoto measure is used to find a measure of dissimilarity between patterns where the features are having binary values. (T)
- 13) K-means algorithm cannot cluster data sets well with large differences in densities. (T)
- 14) Partitional clustering methods help in exploring data at different levels of granularity. (F)

- 15) In case of pattern classification by distance functions, it is always possible to represent each pattern class by a single prototype. (T)
- 16) Kohonen self-organizing network can be used for data coding. (T)
- 17) Density-based clustering algorithms provide a natural protection against outliers.
- 18) In case of pattern classification method using decision function, once a certain function or functions have been selected the only way to establish the effectiveness of the chosen decision function is by direct trial. (T)
- 19) The measurement unit used cannot affect the clustering result of a given data set. (T)
- 20) DBSCAN algorithm cannot cluster data sets well with large differences in densities.
- 21) Hierarchical clustering methods help in exploring data at different levels of granularity. (T)
- 22) A large part of data will not be clustered if a high value of ϵ is chosen in DBSCAN algorithm.
- 23) Data mining engine acts as the domain knowledge that is used to guide the search on to evaluate the interestingness of resulting patterns.
- 24) Hopfield network is usually employed for data clustering. (F)
- 25) A NB classifier is much faster than KNN to provide prediction. (T)
- 26) A Bayesian network is not suitable to represent the probabilistic relationship between diseases and their symptoms. (F)
- 27) A pattern is a shape of an object that we are going to recognize. (F)
- 28) The method of pattern classification by decision function can be expected to yield satisfactory results only when the patterns are linearly separable. (F)